Weighted-Proj integrality: a checked prerequisite and open gluing gate

Bounded formal proof investigation

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The new Lean theorem proves that the infinity chart coordinate $\bar{z}$ is nonzero for every field and coefficient tuple. It does not prove that the whole weighted-Proj curve is integral, construct its global function field, or establish a divisor/Picard or compiled-runtime statement.

Checked result

`WeightedProjIntegral.infinity_chart_z_ne_zero` uses the checked relation $\bar{w}^2 = \bar{z}\bar{u}$ and the checked fact $\bar{w} \neq 0$. Pinned Lean 4.34.1 exited zero; direct axiom output is `[propext, Classical.choice, Quot.sound]`. The exact source and compile output accompany this report.

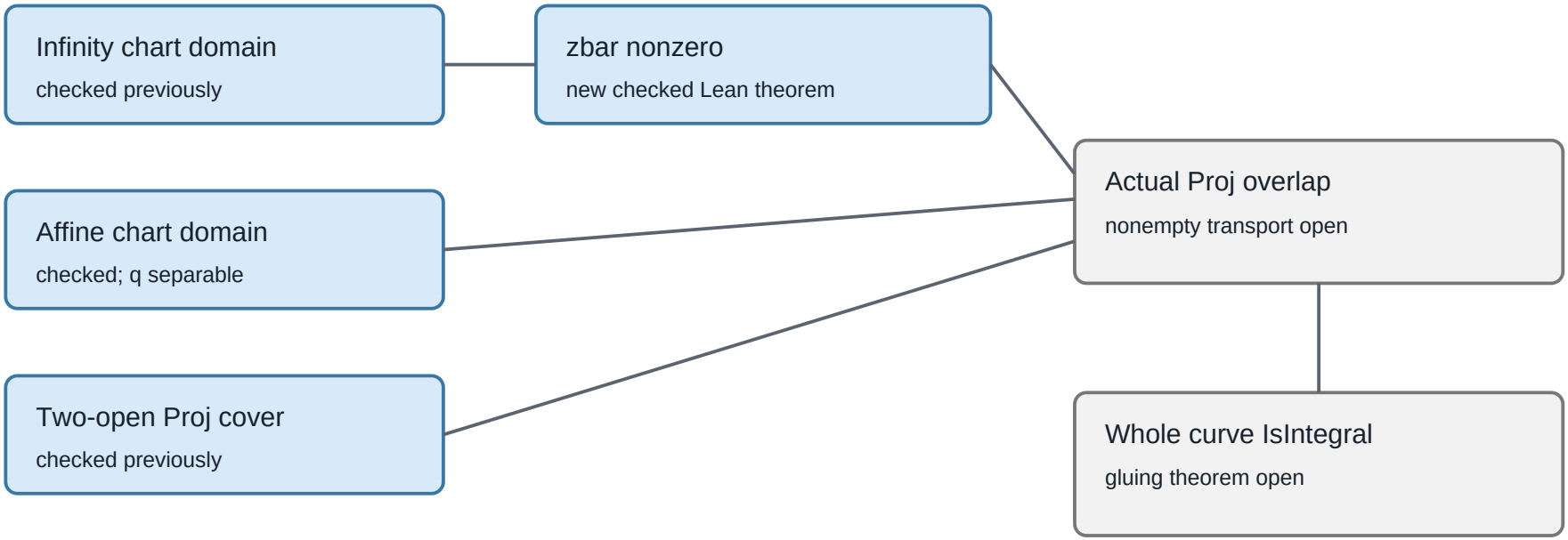
Remaining proof bridge

Existing proofs give a two-basic-open cover by the X and Z opens, an unconditional domain for the infinity chart, and a domain for the affine chart when the quintic is separable. The new nonzero result is a prerequisite for finding a point of $D(\bar{z})$ in the infinity-chart spectrum. The missing checked step transports that point through the chart isomorphism and identifies it as lying in the actual Z-open of weighted Proj. Without a nonempty intersection, two domain opens may form a disconnected reducible union.

After that transport, pinned Mathlib's `PreirreducibleSpace.of_isOpenCover`, `IsReduced.of_openCover`, and `isIntegral_of_irreducibleSpace_of_isReduced` provide a route to the whole-curve instance. This route is a dependency analysis; its conclusion is open. The CSV records the exact status of each step. No DLP experiments or cost measurements were run.

Proof dependencies

Weighted-Proj integrality proof boundary



Blue = kernel-checked declarations. Gray = unproved transport or gluing. Arrows = dependencies.

Blue nodes are checked declarations; gray nodes are open. Arrows represent proof dependencies, not performance measurements.